Customer Churn Analysis — Banking Sector

Project Objective

The objective of this project was to analyze customer churn behavior in a banking dataset and identify the factors influencing customer attrition. The project combines exploratory data analysis, statistical modeling, and machine learning techniques to derive actionable business insights and predict churn probability.

Dataset Overview

The dataset contains customer-level banking information including:

- Demographics
- Geography
- Credit score
- Account balance
- Product holdings
- Customer activity
- Satisfaction metrics
- Churn status

The target variable used in the analysis was:

- **Exited**
  - `1 = Customer Churned`
  - `0 = Customer Retained`

Exploratory Data Analysis (EDA)

Customer Churn Rate

Initial exploration of the dataset showed a measurable proportion of customers exiting the bank, indicating a relevant customer retention problem.

Demographic Insights

Age and Churn

One of the strongest observed trends was the relationship between age and churn behavior.

- The average age of customers who churned was **45 years**
- The average age of customers who stayed was **37 years**

This suggests that older customers are significantly more likely to leave the bank.

Geography-Based Churn Patterns

Churn was disproportionately concentrated among customers from Germany.

Compared to other regions:

- German customers exhibited a significantly higher churn tendency
- France appeared to have comparatively stronger customer retention

This indicates possible regional dissatisfaction, competitive pressure, or product-market mismatch within the German market.

Variables With Minimal Observable Impact

The following variables showed limited observable influence on churn during exploratory analysis:

- Credit Score
- Tenure
- Ownership of Credit Card
- Satisfaction Score

These variables did not exhibit major differences between churned and retained customers at the descriptive level.

Statistical Analysis — Logistic Regression

A logistic regression model was developed using `statsmodels` to identify statistically significant predictors of churn.

Significant Findings

Variables That Reduced Churn Probability

The following variables reduced the likelihood of customer churn:

- Higher Credit Score
- Longer Customer Tenure
- Being an Active Member
- Being Male (relative to female customers)

These variables showed negative relationships with churn probability.

Variables That Increased Churn Probability

The following variables increased churn likelihood:

- Higher Age
- Higher Account Balance
- Being Located in Germany

These variables showed positive relationships with churn probability and statistically significant effects.

Variables Found Statistically Insignificant

At the selected confidence interval, the following variables did not show significant predictive power:

- Number of Products
- Having a Credit Card
- Satisfaction Score
- Reward Points Earned

Predictive Modeling

Logistic Regression Classifier

A predictive churn model was built using Scikit-learn Logistic Regression.

Model Performance

Metric	Score
Accuracy	99.85%
ROC-AUC Score	0.9996

The model achieved extremely high predictive performance.

However, such unusually high scores may indicate:

- Potential overfitting
- Highly separable or simulated data
- Data leakage possibilities

Despite this concern, the model demonstrates strong predictive capability for identifying churn behavior.

Business Insights & Recommendations

1. Improve Customer Retention Through Engagement

Since active members are less likely to churn:

- The bank should increase customer engagement initiatives
- Loyalty programs and activity-based incentives may improve retention
- Personalized communication strategies can increase account usage frequency

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2. Focus on Long-Term Customer Relationships

Longer tenure reduces churn probability.

This suggests:

- Customer stickiness strengthens over time
- Retention strategies should especially target newer customers
- Early-stage onboarding experiences are likely critical

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3. Investigate Older Customer Segments

Older customers are significantly more likely to churn.

This requires deeper qualitative investigation through:

- Focus Group Discussions (FGDs)
- Personal Interviews (PIs)
- Customer satisfaction studies

The bank should identify:

- Service gaps
- Trust issues
- Product relevance problems
- Digital adoption barriers among older customers

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4. High-Balance Customers Are Leaving

Customers with higher balances demonstrated greater churn likelihood.

Possible implications:

- Premium offerings may be insufficient
- Competitors may provide better value to affluent customers
- Wealth-management services may require improvement

If the bank intends to serve higher-income segments, this issue requires immediate strategic attention.

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5. Germany Requires Strategic Attention

German customers displayed substantially higher churn tendencies compared to French customers.

The bank should investigate:

- Local competition
- Product-market fit
- Customer experience differences
- Pricing sensitivity
- Regional operational issues

This may require country-specific retention strategies.

Technical Skills Demonstrated

This project demonstrates capabilities in:

Data Analysis

- Exploratory Data Analysis (EDA)
- Descriptive Statistics
- Data Cleaning
- Feature Interpretation

Statistical Techniques

- Logistic Regression
- Hypothesis Interpretation
- Significance Testing

Machine Learning

- Predictive Modeling
- Classification
- ROC-AUC Evaluation
- Model Performance Assessment

Python Libraries Used

- Pandas
- Matplotlib
- Seaborn
- Statsmodels
- Scikit-learn